

Title: Early Detection and Synthesis of a 12 Lead ECG using Density-Based Clustering and Generative Adversarial Networks

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Cardiovascular disease (CVD) is the leading cause of death worldwide, accounting for one in every five deaths. Early detection of CVD can mitigate up to 80% of fatalities but only half of them are detected in time. Currently, there are few methodologies for the early detection of CVD due to the challenges associated with the diversity in the types of heart diseases, and the uniqueness of an individual's ECG. A promising approach to detecting CVD early is to build a baseline of historical ECGs which can then be used to detect changes or anomalies that occur over time.

Smart devices are one of the best methods for building and analyzing 12 lead historical ECGs efficiently but using current recording techniques they are limited to recording only 1 lead. 12 lead Electrocardiograms are the gold standard for CVD detection and diagnosis but can only be recorded using specialized equipment. Few diseases can be detected using 1 lead underlining the need for more accessible 12 lead ECGs.

This abstract proposes a methodology for the synthesis of a 12-lead ECG and Early Detection of CVD. Through the use of predefined placements on the human body, smart devices can record 3 leads. Placing a smart device on the Left Wrist, Left Thigh, and Right Thigh, Lead 1, Lead 2, and Lead 3 can be recorded. These 3 leads can then be used to synthesize a 12-lead ECG.

Using 12 lead ECGs from PTB-XL, a large publicly available electrocardiography dataset, a Generative Adversarial Network (GAN) is trained to recognize the patterns of electrical activity associated with each lead. Once the GAN has been trained, it is personalized by further training the model using the individual's historical ECGs. The model can then generate a 12-lead ECG using only lead 1, lead 2, and lead 3 recorded from a smart device.

Density-based spatial clustering of applications with noise (DBSCAN) is used for early detection of CVD using the synthesized 12 lead ECG. An individual's historical ECGs are used to create a custom baseline, after which DBSCAN is used to identify changes in the ECG compared to the established baseline. The uniqueness of an individual's ECG is accounted for because their own historical ECG is used to detect changes. Through the use of unsupervised machine learning models, DBSCAN allows any heart disease to be detected without pre-classified data.

This research using GANs and DBSCAN is validated based on its accuracy in generating a 12-lead ECG from three leads and the model's ability to detect changes in an ECG. Using the discriminator to identify real and generated outputs the GAN can be tuned to a high accuracy. This approach provides an easily deployable method for the synthesis of a 12-lead ECG and Early Detection of CVD.

References:

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